Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	McCann, Robert Ph.D.		
Project Title:	Development of a crew-systems concept for vehicle health management in next-generation spacecraft		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Robert.S.McCann@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-0052
Organization Name:	NASA AMES Research Center		
PI Address 1:	MS 262-4		
PI Address 2:			
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	07/01/2003	**End Date:**	07/01/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Fault management on board today's spacecraft is a difficult and time consuming activity that must be improved if NASA is to meet its objectives for the safety and operating efficiency of next generation spacecraft. In the short term (Year 1), we propose to validate an empirical approach to the study of fault management behavior in a part-task simulation of a shuttle cockpit. during ascent. The approach will initially investigate the efficacy of non-invasive techniques, such as eye-movement recording, for measuring changes to fault management performance across different user interfaces. One study will measure an expected positive impact on fault management of a recently completed upgrade to the shuttle cockpit displays; the second study will evaluate the effects of several improvements to the shuttle's current caution and warning system. Assuming these studies validate our approach, the approach will form the empirical basis of a long-term program to develop and evaluate advanced user interface concepts for fault management on next-generation spacecraft. Our focus will be on designing and evaluating interfaces that retire human factors risks associated with the incorporation		

	of advanced information technologies (such as automation) into the fault management process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Non-invasive measurement technique will allow us to quantify human/system performance benefits of next-generation spacecraft human-computer interfaces
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)